

Отчёт по лабораторной работе 3

Анализ трафика в Wireshark

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12 января 2026

Цель работы

Изучить в Wireshark кадры Ethernet и проанализировать PDU протоколов транспортного и прикладного уровней стека TCP/IP.

Анализ параметров подключения

Получение информации о сетевом соединении (ipconfig)

По выводу ipconfig определены ключевые параметры Wi-Fi подключения: - адаптер: Intel(R) Wi-Fi 6E AX211 160MHz; - IPv4: 192.168.1.69, маска: 255.255.255.0; - шлюз / DHCP / DNS: 192.168.1.1; - DHCP включён.

```
Адаптер беспроводной локальной сети Беспроводная сеть:

DNS-суффикс подключения . . . . . :
Описание. . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Физический адрес. . . . . : F8-FE-5E-6F-7B-EC
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да
Локальный IPv6-адрес канала . . . : fe80::da3b:4057:9ef4:1e28%7(Основной)
IPv4-адрес. . . . . : 192.168.1.69(Основной)
Маска подсети . . . . . : 255.255.255.0
Аренда получена. . . . . : 12 января 2026 г. 11:23:30
Срок аренды истекает. . . . . : 13 января 2026 г. 11:23:30
Основной шлюз. . . . . : 192.168.1.1
DHCP-сервер. . . . . : 192.168.1.1
IAID DHCPv6 . . . . . : 133758558
DUID клиента DHCPv6 . . . . . : 00-01-00-01-30-47-94-1A-10-FF-E0-21-D6-B4
DNS-серверы. . . . . : 192.168.1.1
NetBios через TCP/IP. . . . . : Включен
```

Рис. 1: Вывод команды ipconfig

Физический адрес Wi-Fi интерфейса: - **F8-FE-5E-6F-7B-EC**

Интерпретация структуры: - **OUI (F8-FE-5E)** — производитель (Intel); - **NIC part (6F-7B-EC)** — уникальная часть интерфейса.

Первый октет: $F8 = 11111000_2$ - I/G (младший бит) = 0 → **unicast** - U/L (второй младший бит) = 0
→ глобально администрируемый

ARP и ICMP в Wireshark

Проверка доступности шлюза (ping 192.168.1.1)

```
PS C:\Users\akabrikosov\Documents> ping 192.168.1.1
```

```
Обмен пакетами с 192.168.1.1 по с 32 байтами данных:
```

```
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
```

```
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
```

```
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
```

```
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
```

```
Статистика Ping для 192.168.1.1:
```

```
Пакетов: отправлено = 4, получено = 4, потеряно = 0  
(0% потерь)
```

```
Приблизительное время приема-передачи в мс:
```

```
Минимальное = 0мсек, Максимальное = 0 мсек, Среднее = 0 мсек
```

```
PS C:\Users\akabrikosov\Documents> |
```

Рис. 2: Проверка доступности шлюза по умолчанию

ICMP echo request (в Wireshark)

Фильтр: arp or icmp

Характеристики кадра: - Ethernet II, 74 байта - SRC MAC: F8-FE-5E-6F-7B-EC - DST MAC: C8-7F-54-78-B6-F2

No.	Time	Source	Destination	Protocol	Length	Info
76	51.216617	ASUSTekCOMPU_78:b6:f2...	Intel_6f:7b:ec	ARP	46	Who has 192.168.1.69? Tell 192.168.1.1
77	51.216649	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
→ 103	72.435246	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=1/256,
← 104	72.435808	192.168.1.1	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=1/256,
106	73.441009	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2/512,
107	73.441761	192.168.1.1	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=2/512,
108	74.453620	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=3/768,

> Frame 103: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{3ABB4A...	0000	c8
▼ Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0010	00
▼ Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0020	01
.... 0. = LG bit: Globally unique address (factory default)	0030	67
.... 0. = IG bit: Individual address (unicast)	0040	77
▼ Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)		
.... 0. = LG bit: Globally unique address (factory default)		
.... 0. = IG bit: Individual address (unicast)		
Type: IPv4 (0x0800)		
[Stream index: 1]		
> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 192.168.1.1		
> Internet Control Message Protocol		

ICMP echo reply (в Wireshark)

Характеристики кадра: - Ethernet II, 78 байт - SRC MAC: C8-7F-54-78-B6-F2 - DST MAC: F8-FE-5E-6F-7B-EC

arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
76	51.216617	ASUSTekCOMPU_78:b6:f2...	Intel_6f:7b:ec	ARP	46	Who has 192.168.1.69? Tell 192.168.1.69
77	51.216649	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
→ 103	72.435246	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=0x00000001
← 104	72.435808	192.168.1.1	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=0x00000001
106	73.441009	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=0x00000001
107	73.441761	192.168.1.1	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=0x00000001
108	74.453620	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=0x00000001

Frame 104: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{3ABB4A...}

- Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
 - Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)
 - Source: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)
 - Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 1]
- 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
- Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.69
- Internet Control Message Protocol

ARP-запрос и ARP-ответ

ARP-запрос: - Src IP: 192.168.1.69, Dst IP: 192.168.1.1 - Dst MAC: broadcast

ARP-ответ: - Src IP: 192.168.1.1 - Dst MAC: MAC компьютера

No.	Time	Source	Destination	Protocol	Length	Info
7	6.161835	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	46	Who has 192.168.1.69? Tell 192.168.1.1
8	6.161864	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
14	12.408015	e6:2b:fe:93:3e:82	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.26
76	51.216617	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	46	Who has 192.168.1.69? Tell 192.168.1.1
77	51.216649	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
103	72.435246	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, 1
104	72.435808	192.168.1.1	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=1/256, 1

```
> Frame 8: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{3ABB4A4F
▼ Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
  ▼ Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  ▼ Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 1]
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
  Sender IP address: 192.168.1.69
```

```
0000 c8 7f 54 78 b6 f2
0010 08 06
0020 c8 7f 54 78 b6 f2
```

DNS и внешний доступ

Проверка доменного имени (ping yandex.ru)

Получение ICMP ответов подтверждает: - корректную работу DNS-резолвинга; - доступ в сеть через шлюз.

```
PS C:\Users\akabrikosov\Documents> ping yandex.ru

Обмен пакетами с yandex.ru [77.88.44.55] с 32 байтами данных:
Ответ от 77.88.44.55: число байт=32 время=4мс TTL=53
Ответ от 77.88.44.55: число байт=32 время=5мс TTL=53
Ответ от 77.88.44.55: число байт=32 время=5мс TTL=53
Ответ от 77.88.44.55: число байт=32 время=5мс TTL=53

Статистика Ping для 77.88.44.55:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)

Приблизительное время приема-передачи в мс:
    Минимальное = 4мсек, Максимальное = 5 мсек, Среднее = 4 мсек
PS C:\Users\akabrikosov\Documents> |
```

Рис. 6: Ping доменного имени yandex.ru

Пакеты при обращении к внешнему ресурсу

На канальном уровне трафик идёт: - SRC MAC: MAC компьютера - DST MAC: MAC шлюза

No.	Time	Source	Destination	Protocol	Length	Info
15	11.741233	77.88.44.55	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=7/1792,
17	12.742712	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048,
18	12.747817	77.88.44.55	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=8/2048,
20	14.420369	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42	Who has 192.168.1.1? Tell 192.168.1.69
21	14.421217	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	46	192.168.1.1 is at c8:7f:54:78:b6:f2
22	14.916706	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	46	Who has 192.168.1.69? Tell 192.168.1.1
23	14.916757	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec

```
> Frame 20: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{3ABB4A4...}
Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
  Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0 .... = IG bit: Individual address (unicast)
  Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0 .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 3]
Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
  Sender IP address: 192.168.1.69
  Target MAC address: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
  Target IP address: 192.168.1.1
```

```
0000 c8 7f
0010 08 06
0020 c8 7f
```

Заголовок IPv4 (пример ICMP)

Параметры заголовка IPv4: - IHL: 20 байт - TTL: 128 - Protocol: ICMP

No.	Time	Source	Destination	Protocol	Length	Info
9	9.717800	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280,
10	9.722740	77.88.44.55	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=5/1280,
12	10.725329	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536,
13	10.730770	77.88.44.55	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=6/1536,
14	11.736135	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792,
15	11.741233	77.88.44.55	192.168.1.69	ICMP	78	Echo (ping) reply id=0x0001, seq=7/1792,
17	12.742712	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048,

> Frame 9: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{3ABB4A...}	0000	c8 7f
▼ Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0010	00 3c
▼ Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0020	2c 37
.... 0. = LG bit: Globally unique address (factory default)	0030	67 68
.... 0. = IG bit: Individual address (unicast)	0040	77 61
▼ Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)		
.... 0. = LG bit: Globally unique address (factory default)		
.... 0. = IG bit: Individual address (unicast)		
Type: IPv4 (0x0800)		
[Stream index: 3]		
▼ Internet Protocol Version 4, Src: 192.168.1.69, Dst: 77.88.44.55		
0100 = Version: 4		
.... 0101 = Header Length: 20 bytes (5)		
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)		
Total Length: 60		
Identification: 0x6cb9 (27833)		
> 000. = Flags: 0x0		
...0 0000 0000 0000 = Fragment Offset: 0		
Time to Live: 128		
Protocol: ICMP (1)		
Header Checksum: 0x0000 [validation disabled]		
[Header checksum status: Unverified]		
Source Address: 192.168.1.69		

Транспортный и прикладной уровни

HTTP поверх TCP (порт 80): запрос

Фильтр: http

Зафиксировано: - Src IP: 192.168.1.69 - Dst IP: 188.184.67.127 - TCP Src port: 52480, Dst port: 80 -

Flags: PSH, ACK - Payload: 448 байт, TCP header: 20 байт

132	5.1813/6	88.221.132.90	192.168.1.69	HTTP	322 HTTP/1.1 304 Not Modified
181	28.443115	192.168.1.69	188.184.67.127	HTTP	502 GET / HTTP/1.1
183	28.493145	188.184.67.127	192.168.1.69	HTTP	932 HTTP/1.1 200 OK (text/html)
187	28.511752	192.168.1.69	188.184.67.127	HTTP	443 GET /favicon.ico HTTP/1.1
191	28.562695	188.184.67.127	192.168.1.69	HTTP	248 HTTP/1.1 200 OK (image/vnd.microsoft.icon)
208	31.640081	192.168.1.69	188.184.67.127	HTTP	562 GET /hypertext/WWW/TheProject.html HTTP/1.1
221	31.691170	188.184.67.127	192.168.1.69	HTTP	1044 HTTP/1.1 200 OK (text/html)

> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0000	c8
> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 188.184.67.127	0010	01
▼ Transmission Control Protocol, Src Port: 52480, Dst Port: 80, Seq: 1, Ack: 1, Len: 448	0020	43
Source Port: 52480	0030	02
Destination Port: 80	0040	2f
[Stream index: 1]	0050	2e
[Stream Packet Number: 4]	0060	74
> [Conversation completeness: Complete, WITH_DATA (31)]	0070	0d
[TCP Segment Len: 448]	0080	72
Sequence Number: 1 (relative sequence number)	0090	55
Sequence Number (raw): 4007524945	00a0	6c
[Next Sequence Number: 449 (relative sequence number)]	00b0	20
Acknowledgment Number: 1 (relative ack number)	00c0	20
Acknowledgment number (raw): 4219280799	00d0	74
0101 = Header Length: 20 bytes (5)	00e0	20
> Flags: 0x018 (PSH, ACK)	00f0	6f
Window: 513	0100	66
[Calculated window size: 131328]	0110	65
	0120	70
	0130	2b
	0140	2f

HTTP поверх TCP (порт 80): ответ

Сервер отвечает по тому же соединению: - Src port: 80 → Dst port: 52480 - Flags: PSH, ACK -
Payload: 878 байт, TCP header: 20 байт

No.	Time	Source	Destination	Protocol	Length	Info
132	5.181376	88.221.132.90	192.168.1.69	HTTP	322	HTTP/1.1 304 Not Modified
181	28.443115	192.168.1.69	188.184.67.127	HTTP	502	GET / HTTP/1.1
183	28.493145	188.184.67.127	192.168.1.69	HTTP	932	HTTP/1.1 200 OK (text/html)
187	28.511752	192.168.1.69	188.184.67.127	HTTP	443	GET /favicon.ico HTTP/1.1
191	28.562695	188.184.67.127	192.168.1.69	HTTP	248	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
208	31.640081	192.168.1.69	188.184.67.127	HTTP	562	GET /hypertext/WWW/TheProject.html HTTP/1.1
221	31.691170	188.184.67.127	192.168.1.69	HTTP	1044	HTTP/1.1 200 OK (text/html)

> Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0000	f8
> Internet Protocol Version 4, Src: 188.184.67.127, Dst: 192.168.1.69	0010	03
▼ Transmission Control Protocol, Src Port: 80, Dst Port: 52480, Seq: 1, Ack: 449, Len: 878	0020	01
Source Port: 80	0030	01
Destination Port: 52480	0040	30
[Stream index: 1]	0050	2c
[Stream Packet Number: 6]	0060	3a
> [Conversation completeness: Complete, WITH_DATA (31)]	0070	65
[TCP Segment Len: 878]	0080	2d
Sequence Number: 1 (relative sequence number)	0090	30
Sequence Number (raw): 4219280799	00a0	30
[Next Sequence Number: 879 (relative sequence number)]	00b0	22
Acknowledgment Number: 449 (relative ack number)	00c0	63
Acknowledgment number (raw): 4007525393	00d0	65
0101 = Header Length: 20 bytes (5)	00e0	6e
> Flags: 0x018 (PSH, ACK)	00f0	43
Window: 501	0100	65
[Calculated window size: 64128]	0110	20
[Window size scaling factor: 128]	0120	74
	0130	3e
	0140	3c

DNS поверх UDP: запрос

Фильтр: dns

Параметры: - Src IP: 192.168.1.69 → Dst IP: 192.168.1.1 - Src port: 63313 → Dst port: 53 - UDP
payload: 35 байт - Transaction ID: 0x1ed8

No.	Time	Source	Destination	Protocol	Length	Info
122	5.125056	192.168.1.1	192.168.1.69	DNS	121	Standard query response 0xb58f A c.pki.gov
148	20.258078	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x03e1 HTTPS suggest.yande
149	20.258228	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x1ed8 A suggest.yandex.ru
150	20.263719	192.168.1.1	192.168.1.69	DNS	160	Standard query response 0x03e1 HTTPS sugge
151	20.268849	192.168.1.1	192.168.1.69	DNS	125	Standard query response 0x1ed8 A suggest.y
197	30.682112	192.168.1.69	192.168.1.1	DNS	77	Standard query 0xc07f HTTPS line-mode.cerr
198	30.682284	192.168.1.69	192.168.1.1	DNS	77	Standard query 0xf6da A line-mode.cern.ch

> Frame 149: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \Device\NPF_{3ABB4A}	0000	c8
> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0010	00
> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 192.168.1.1	0020	01
✓ User Datagram Protocol, Src Port: 63313, Dst Port: 53	0030	06
Source Port: 63313	0040	61
Destination Port: 53		
Length: 43		
Checksum: 0x83d3 [unverified]		
[Checksum Status: Unverified]		
[Stream index: 14]		
[Stream Packet Number: 1]		
> [Timestamps]		
UDP payload (35 bytes)		
✓ Domain Name System (query)		
Transaction ID: 0x1ed8		

DNS поверх UDP: ответ

Параметры ответа: - Src IP: 192.168.1.1 → Dst IP: 192.168.1.69 - Transaction ID: 0x1ed8 (совпадает) -
No error, Answer RRs: 2

dns						
No.	Time	Source	Destination	Protocol	Length	Info
148	20.258078	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x03e1 HTTPS suggest.yandex.
149	20.258228	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x1ed8 A suggest.yandex.ru
150	20.263719	192.168.1.1	192.168.1.69	DNS	160	Standard query response 0x03e1 HTTPS sugges
151	20.268849	192.168.1.1	192.168.1.69	DNS	125	Standard query response 0x1ed8 A suggest.ya
197	30.682112	192.168.1.69	192.168.1.1	DNS	77	Standard query 0xc07f HTTPS line-mode.cern.
198	30.682284	192.168.1.69	192.168.1.1	DNS	77	Standard query 0xf6da A line-mode.cern.ch
199	30.790071	192.168.1.1	192.168.1.69	DNS	157	Standard query response 0xf6da A line-mode.

> Frame 151: Packet, 125 bytes on wire (1000 bits), 125 bytes captured (1000 bits) on interface \Device\NPF_{		0000	f8
> Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)		0010	00
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.69		0020	01
User Datagram Protocol, Src Port: 53, Dst Port: 63313		0030	00
Source Port: 53		0040	61
Destination Port: 63313		0050	05
Length: 91		0060	74
Checksum: 0xfba2 [unverified]		0070	01
[Checksum Status: Unverified]			
[Stream index: 14]			
[Stream Packet Number: 2]			
> [Timestamps]			
UDP payload (83 bytes)			
Domain Name System (response)			
Transaction ID: 0x1ed8			
> Flags: 0x8180 Standard query response, No error			
Questions: 1			

QUIC поверх UDP: Initial (к серверу)

Фильтр: quic

Ключевые поля: - Long Header, Packet Type: Initial - Version: 1 (0x00000001) - DCID Length: 8 - DCID: 77058adeeee4b870c

The image shows a Wireshark packet capture window titled "Захват с Беспроводная сеть". The filter bar at the top is set to "quic". The packet list shows several packets, with packet 32 selected. The packet details pane shows the structure of the QUIC Initial packet, including the Long Header, Fixed Bit, Packet Type, Version, and Connection IDs.

No.	Time	Source	Destination	Protocol	Length	Info
32	2.489633	192.168.1.69	146.75.121.91	QUIC	1292	Initial, DCID=77058adeeee4b870c, PKN: 1, CRYPTO...
33	2.489670	192.168.1.69	146.75.121.91	QUIC	1292	Initial, DCID=77058adeeee4b870c, PKN: 2, PING...
34	2.490261	192.168.1.69	146.75.121.91	QUIC	117	0-RTT, DCID=77058adeeee4b870c
35	2.529252	146.75.121.91	192.168.1.69	QUIC	1322	Initial, SCID=f6b296f219f55030078ddc1e885930...
36	2.529918	146.75.121.91	192.168.1.69	QUIC	1322	Initial, SCID=f6b296f219f55030078ddc1e885930...
37	2.529918	146.75.121.91	192.168.1.69	QUIC	637	Protected Payload (KP0)
38	2.530178	192.168.1.69	146.75.121.91	QUIC	145	Handshake, DCID=f6b296f219f55030078ddc1e8859...

UDP payload (1250 bytes)

- QUIC IETF
 - QUIC Connection information
 - [Packet Length: 1250]
 - 1... .. = Header Form: Long Header (1)
 - .1.. = Fixed Bit: True
 - ..00 = Packet Type: Initial (0)
 - [.... 00.. = Reserved: 0]
 - [.... ..00 = Packet Number Length: 1 bytes (0)]
 - Version: 1 (0x00000001)
 - Destination Connection ID Length: 8
 - Destination Connection ID: 77058adeeee4b870c
 - Source Connection ID Length: 0
 - Token Length: 0

Hex dump (right side of the packet details pane):

```
0020 79 50 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030 77 05 8a de ee e4 b8 70 c0 00 00 00 00 00 00 00
0040 cd 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050 9d 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 cf 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 fd 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 82 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 bb 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 cc 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 a9 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 16 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 cd 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 30 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 64 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

QUIC: Initial (ответ сервера)

Параметры: - Packet Type: Initial - SCID Length: 17 - SCID: f6b296f219f55030078ddc1e885930e148

The image shows a Wireshark packet capture of a QUIC Initial packet. The packet list pane at the top shows several packets, with packet 35 selected. The packet details pane below shows the structure of the selected packet, including the QUIC IETF section with connection information and the payload.

No.	Time	Source	Destination	Protocol	Length	Info
32	2.489633	192.168.1.69	146.75.121.91	QUIC	1292	Initial, DCID=77058adeee4b870c, PKN: 1, CRYPT
33	2.489670	192.168.1.69	146.75.121.91	QUIC	1292	Initial, DCID=77058adeee4b870c, PKN: 2, PING
34	2.490261	192.168.1.69	146.75.121.91	QUIC	117	0-RTT, DCID=77058adeee4b870c
35	2.529252	146.75.121.91	192.168.1.69	QUIC	1322	Initial, SCID=f6b296f219f55030078ddc1e885930
36	2.529918	146.75.121.91	192.168.1.69	QUIC	1322	Initial, SCID=f6b296f219f55030078ddc1e885930
37	2.529918	146.75.121.91	192.168.1.69	QUIC	637	Protected Payload (KP0)
38	2.530178	192.168.1.69	146.75.121.91	QUIC	145	Handshake, DCID=f6b296f219f55030078ddc1e8859

Packet 35 details:

- [Stream index: 8]
- [Stream Packet Number: 4]
- > [Timestamps]
- UDP payload (1280 bytes)
- QUIC IETF
 - > QUIC Connection information
 - [Packet Length: 1280]
 - 1... = Header Form: Long Header (1)
 - .1.. = Fixed Bit: True
 - ..00 = Packet Type: Initial (0)
 - [.... 00.. = Reserved: 0]
 - [.... ..01 = Packet Number Length: 2 bytes (1)]
 - Version: 1 (0x00000001)
 - Destination Connection ID Length: 0
 - Source Connection ID Length: 17
 - Source Connection ID: f6b296f219f55030078ddc1e885930e148
 - Token Length: 0
 - Length: 1253
 - [Packet Number: 0]
 - Payload [...]: de8ed48e8c57c94f2835af66fc9e3a50fe34b921a374644f05dcb34390afabc1f11b63cebbc91663efda5148b0

Hex dump (offset 0020):

```
0020 01 4
0030 11 4
0040 e1 4
0050 35 4
0060 5d 4
0070 fd 4
0080 94 4
0090 a0 4
00a0 05 4
00b0 3d 4
00c0 43 4
00d0 e6 4
00e0 0c 4
00f0 0d 4
0100 8f 4
0110 60 4
0120 16 4
0130 1a 4
0140 26 4
0150 5b 4
0160 5f 4
0170 08 4
0180 ae 4
```

TCP handshake

В Wireshark отфильтрованы TCP пакеты и выбран поток между: - 192.168.1.69 ↔ 188.184.67.127

13	8.364706	192.168.1.69	213.180.204.63	TCP	54 62467 → 443 [RST] Seq=3 Win=0 Len=0
14	8.408512	188.184.67.127	192.168.1.69	TCP	66 80 → 51617 [SYN, ACK] Seq=0 Ack=1 Win=64240
15	8.408673	192.168.1.69	188.184.67.127	TCP	54 51617 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=
16	8.462305	192.168.1.69	188.184.67.127	TCP	66 50071 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=
17	8.462445	192.168.1.69	188.184.67.127	HTTP	592 GET /hypertext/DataSources/Top.html HTTP/1.
25	8.511380	188.184.67.127	192.168.1.69	TCP	66 80 → 50071 [SYN, ACK] Seq=0 Ack=1 Win=64240
26	8.511431	192.168.1.69	188.184.67.127	TCP	54 50071 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=
27	8.512484	188.184.67.127	192.168.1.69	TCP	54 80 → 51617 [ACK] Seq=1 Ack=539 Win=63744 Len
28	8.514262	188.184.67.127	192.168.1.69	HTTP	950 HTTP/1.1 200 OK (text/html)
29	8.514262	188.184.67.127	192.168.1.69	TCP	54 80 → 51617 [FIN, ACK] Seq=897 Ack=539 Win=6:
30	8.514284	192.168.1.69	188.184.67.127	TCP	54 51617 → 80 [ACK] Seq=539 Ack=898 Win=130304
31	8.514624	192.168.1.69	188.184.67.127	TCP	54 51617 → 80 [FIN, ACK] Seq=539 Ack=898 Win=1:
38	8.563865	188.184.67.127	192.168.1.69	TCP	54 80 → 51617 [ACK] Seq=898 Ack=540 Win=63744 I
47	13.200738	192.168.1.69	64.233.161.188	TCP	55 64117 → 5228 [ACK] Seq=1 Ack=1 Win=512 Len=
48	13.221347	64.233.161.188	192.168.1.69	TCP	66 5228 → 64117 [ACK] Seq=1 Ack=2 Win=1046 Len=

Рис. 15: Общий список TCP-пакетов и выбранный поток

Этап 1: SYN (клиент → сервер)

Клиент инициирует соединение: - Src port: **50071** → Dst port: **80** - Flags: **SYN** - Seq: **0** - TCP header: **32 байта** (есть опции)

16	8.462305	192.168.1.69	188.184.67.127	TCP	66 50071 → 80 [SYN] Seq=0 Win=64240 Len=0 M
17	8.462445	192.168.1.69	188.184.67.127	HTTP	592 GET /hypertext/DataSources/Top.html HTTP,
25	8.511380	188.184.67.127	192.168.1.69	TCP	66 80 → 50071 [SYN, ACK] Seq=0 Ack=1 Win=64;
26	8.511431	192.168.1.69	188.184.67.127	TCP	54 50071 → 80 [ACK] Seq=1 Ack=1 Win=131328
27	8.512484	188.184.67.127	192.168.1.69	TCP	54 80 → 51617 [ACK] Seq=1 Ack=539 Win=63744
28	8.514262	188.184.67.127	192.168.1.69	HTTP	950 HTTP/1.1 200 OK (text/html)

Destination Port: 80	0000
[Stream index: 2]	0010
[Stream Packet Number: 1]	0020
> [Conversation completeness: Incomplete, DATA (15)]	0030
[TCP Segment Len: 0]	0040
Sequence Number: 0 (relative sequence number)	
Sequence Number (raw): 705851561	
[Next Sequence Number: 1 (relative sequence number)]	
Acknowledgment Number: 0	
Acknowledgment number (raw): 0	
1000 = Header Length: 32 bytes (8)	
> Flags: 0x002 (SYN)	
Window: 64240	
[Calculated window size: 64240]	
Checksum: 0xc24b [unverified]	
[Checksum Status: Unverified]	
Urgent Pointer: 0	
> Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP)	
> [Timestamps]	
[Client Contiguous Streams: 1]	
[Server Contiguous Streams: 1]	

Этап 2-3: SYN/ACK и ACK

SYN/ACK (сервер → клиент): - Flags: SYN, ACK - Ack: 1, Window: 63744

ACK (клиент → сервер): - Flags: ACK - Seq: 1, Ack: 1

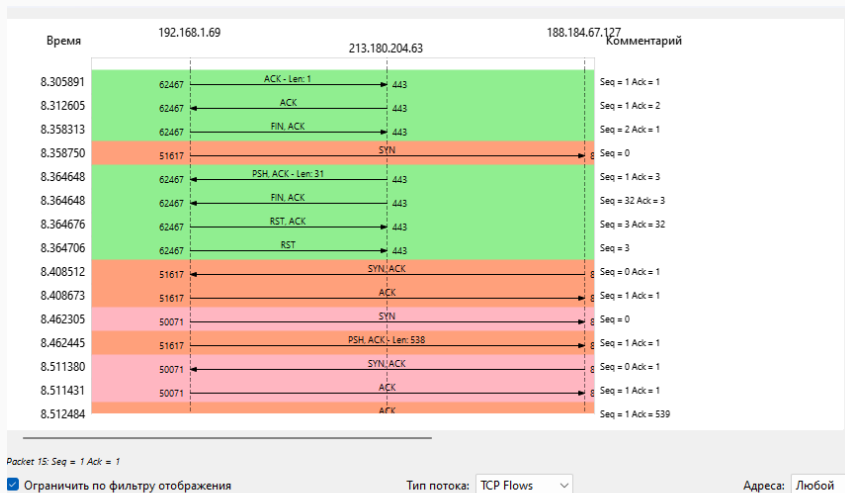
No.	Time	Source	Destination	Protocol	Length	Info
15	8.408673	192.168.1.69	188.184.67.127	TCP	54	51617 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=
16	8.462305	192.168.1.69	188.184.67.127	TCP	66	50071 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1
17	8.462445	192.168.1.69	188.184.67.127	HTTP	592	GET /hypertext/DataSources/Top.html HTTP/1.1
25	8.511380	188.184.67.127	192.168.1.69	TCP	66	80 → 50071 [SYN, ACK] Seq=0 Ack=1 Win=64240
26	8.511431	192.168.1.69	188.184.67.127	TCP	54	50071 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=
27	8.512484	188.184.67.127	192.168.1.69	TCP	54	80 → 51617 [ACK] Seq=1 Ack=539 Win=63744 Len=
28	8.514262	188.184.67.127	192.168.1.69	HTTP	950	HTTP/1.1 200 OK (text/html)
29	8.514262	188.184.67.127	192.168.1.69	TCP	54	80 → 51617 [FIN, ACK] Seq=897 Ack=539 Win=63

Destination Port: 51617
[Stream index: 1]
[Stream Packet Number: 6]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 896]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 481293145
[Next Sequence Number: 897 (relative sequence number)]
Acknowledgment Number: 539 (relative ack number)
Acknowledgment number (raw): 4183877899
0101 = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
Window: 498
[Calculated window size: 63744]
[Window size scaling factor: 128]
Checksum: 0xbff1 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0

0000 f8 1
0010 03 a
0020 01 4
0030 01 1
0040 30 5
0050 2c 2
0060 3a 5
0070 65 7
0080 2d 4
0090 32 5
00a0 30 5
00b0 22 5
00c0 34 5
00d0 65 7
00e0 6e 7
00f0 43 6
0100 65 6
0110 20 7
0120 69 7
0130 20 7
0140 8e 7

Завершение соединения и график потока

График потока визуализирует: - SYN → SYN/ACK → ACK, - передачу данных, - завершение сеанса.



Итоги

В ходе работы: - определены параметры сетевого подключения (IP, шлюз, DHCP/DNS, MAC); - изучены ARP и ICMP на канальном/сетевом уровнях; - проанализированы HTTP/TCP, DNS/UDP, QUIC/UDP; - разобран механизм **TCP three-way handshake** и завершение соединения.

Результаты подтверждают корректную работу сетевого стека Windows и типовой обмен данными в локальной сети и при доступе к внешним ресурсам.